

Econ 440: Unconfoundedness-Based Design: Estimation

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1 Introduction

In these notes, we discuss estimation of the ATE and the ATT in an unconfoundedness-based design. In other words, we ask, how can we estimate the ATE and the ATT in this design. We start by discussing estimation of the ATE , and later we discuss estimation of the ATT .

Remember that the two assumptions on which unconfoundedness-based designs are based are Unconfoundedness (Assumption 1) and Overlap (Assumption 2). As discussed in detail in the previous notes, under these two assumptions we can identify the ATE (i.e., write it as a function of *observed* variables) as:

$$ATE = E_X[E[Y|T = 1, X = x] - E[Y|T = 0, X = x]] \quad (1)$$

If we only have *one* binary X (e.g., gender), then it is easy to estimate the ATE based on equation (1) by using an approach similar to what we did for an experimental design for *each* of the values of X (see last lecture).

Example. Suppose unconfoundedness holds with one covariate, gender, with $X = 1$ if the individual is female, and $X = 0$ if the individual is male. Under unconfoundedness, it is “as if” we have an experiment within the $X = 1$ group (females) and the $X = 0$ group (males).

Hence, within the $X = x$ group we can identify the ATE for that group as:

$$ATE(x) = \underbrace{E[Y|T = 1, X = x]}_{\text{Mean Outcome for Treated}} - \underbrace{E[Y|T = 0, X = x]}_{\text{Mean Outcome for Control}} \quad (2)$$

Therefore, for one binary X we get:

$$ATE = \Pr(X = 1)ATE(X = 1) + \Pr(X = 0)ATE(X = 0)$$

Challenge: In order to argue that unconfoundedness holds, we need to adjust for *all* confounders. This requires us to often control for many X 's (some of which may be continuous) $\implies$ The approach in the example above is not applicable in general.

Some tools or methods to estimate the ATE that we will study in this class include: Ordinary Least Squares (OLS), matching, and propensity score methods.

In the present notes, we focus on OLS.

2 Estimaion of the ATE Based on OLS

2.1 Brief OLS Review

Suppose that we have K covariates $(X_1, X_2, \dots, X_K)$. Recall that OLS makes the assumption that the conditional mean of the outcome Y has the following linear and additive form as a function of the covariates:

$$E[Y|X] = \beta_0 + \beta_1 X_1 + \dots + \beta_K X_K$$

Or,

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_K X_K + \varepsilon,$$

with $E[\varepsilon] = E[\varepsilon|X] = 0$.

2.2 Using OLS to Estimate the ATE

To estimate the ATE using OLS, we first use the Law of Iterated Expectations to write the ATE as:

$$ATE = E[Y(1) - Y(0)]$$

$$ATE = E_X[E[Y(1) - Y(0)|X = x]]$$

$$\implies ATE = E_X[\underbrace{E[Y(1)|X = x] - E[Y(0)|X = x]}_{=ATE(x)}] \quad (3)$$

We base estimation of the *ATE* using OLS on equation (3). In addition to Unconfoundedness (Assumption 1) and Overlap (Assumption 2), OLS imposes the following assumption.

Assumption 3 (Linearity of the Conditional Means of $Y(0)$ and $Y(1)$). Assume that the conditional means $E[Y(0)|X = x]$ and $E[Y(1)|X = x]$ are *Linear* and *Additive* functions of X . That is, assume that:

$$E[Y(0)|X = x] = \alpha_0 + \beta_{01}X_1 + \beta_{02}X_2 + \dots + \beta_{0K}X_K, \text{ and} \quad (4)$$

$$E[Y(1)|X = x] = \alpha_1 + \beta_{11}X_1 + \beta_{12}X_2 + \dots + \beta_{1K}X_K \quad (5)$$

Notes:

- Note that Assumption 3 is imposed on the *population or true* expectations.

- Another way to write equations (4) and (5) is:

$$Y(0) = \alpha_0 + \beta_{01}X_1 + \beta_{02}X_2 + \dots + \beta_{0K}X_K + \varepsilon_0, \text{ and} \quad (6)$$

$$Y(1) = \alpha_1 + \beta_{11}X_1 + \beta_{12}X_2 + \dots + \beta_{1K}X_K + \varepsilon_1 \quad (7)$$

with $E[\varepsilon_0] = E[\varepsilon_0|X] = 0$ and $E[\varepsilon_1] = E[\varepsilon_1|X] = 0$.

- This is another *huge assumption*: We assume a *linear* form for the *conditional means* of the potential outcomes.
 - In principle, there is no reason why the mean potential outcomes (e.g., earnings under attending college, or $Y(1)$) should be a linear and additive function of the covariates (e.g., age, I.Q., gender, parents' income, etc.).
 - As usual, we can think of equations (4) and (5) as *linear approximations* to the true or actual conditional means. However, it is important to keep in mind that linear approximations may work well in certain settings, but (very) bad in others!
 - As we will see later, Assumption 3 is not needed (i.e., necessary) to estimate the *ATE* in unconfoundedness-based designs. Other methods do not impose this assumption (e.g., matching methods).

Under unconfoundedness, we have that: $T \perp\!\!\!\perp \{Y(0), Y(1)\} | X$. Therefore, we can estimate the two regression functions in equations (4) and (5) as follows.

- Consider $E[Y(0)|X = x]$. By unconfoundedness, we have:

$$E[Y(0)|X = x] = E[Y(0)|T = 0, X = x] = E[Y|T = 0, X = x] \quad (8)$$

Therefore, we can estimate $E[Y(0)|X = x]$ by running a linear regression (OLS) of Y on X for those with $T = 0$.

- Next, consider $E[Y(1)|X = x]$. By unconfoundedness, we have:

$$E[Y(1)|X = x] = E[Y(1)|T = 1, X = x] = E[Y|T = 1, X = x] \quad (9)$$

Hence, we can estimate $E[Y(1)|X = x]$ by running an OLS of Y on X for those with $T = 1$.

Note that if we plug the results in equations (8) and (9) into equation (3), we obtain the identification result in our previous notes, which stated that under unconfoundedness and overlap we can write the ATE as (see also equation (1) in the present notes):

$$ATE = E_X[E[Y|T = 1, X = x] - E[Y|T = 0, X = x]]$$

When estimating the ATE using OLS, the key insight is that we are now adding the assumption that the conditional means $E[Y|T = 1, X = x]$ (which equals $E[Y(1)|X = x]$ under unconfoundedness) and $E[Y|T = 0, X = x]$ (which equals $E[Y(0)|X = x]$ under unconfoundedness) have a linear form (Assumption 3).

2.2.1 Five-Step Approach to Estimate the ATE Using OLS

In what follows, we assume that we have a random sample size n from our population of interest. Based on the discussion above, *under Assumptions 1 (Unconfoundedness), 2 (Overlap), and 3 (Linearity of the Conditional Means of $Y(0)$ and $Y(1)$)*, we can estimate the ATE using the following five-step procedure:

1. Run OLS of Y on X using units in the control group ($T = 0$).

- Here, we get estimates of $\alpha_0, \beta_{01}, \dots, \beta_{0K}$ in equation (4).

- Intuition: We are estimating the conditional mean of the potential outcome $Y(0)$, $E[Y(0)|X = x]$ in equation (4), using OLS:

$$E[Y(0)|X = x] = \alpha_0 + \beta_{01}X_1 + \beta_{02}X_2 + \dots + \beta_{0K}X_K$$

2. Get predicted values for *all* individuals in the sample (i.e., individuals in both $T = 1$ and $T = 0$ groups). Let $\hat{E}[Y_i(0)|X = X_i]$ denote the predicted value for individual i .

- Get $\hat{E}[Y_i(0)|X = X_i]$ for all i :

$$\hat{E}[Y_i(0)|X = X_i] = \hat{\alpha}_0 + \hat{\beta}_{01}X_{1i} + \hat{\beta}_{02}X_{2i} + \dots + \hat{\beta}_{0K}X_{Ki}$$

- Intuition: We estimate the conditional mean $E[Y(0)|X = X_i]$ for *all* individuals in our sample (controls and treated), since we want to estimate the *ATE*.

3. Run OLS of Y on X using units in the treatment group ($T = 1$).

- Here, we get estimates of $\alpha_1, \beta_{11}, \dots, \beta_{1K}$ in equation (5).

- Intuition: We are estimating the conditional mean of the potential outcome $Y(1)$, $E[Y(1)|X = x]$ in equation (5), using OLS:

$$E[Y(1)|X = x] = \alpha_1 + \beta_{11}X_1 + \beta_{12}X_2 + \dots + \beta_{1K}X_K$$

4. Get predicted values for *all* individuals in the sample (i.e., individuals in both $T = 1, T = 0$ groups). Let $\hat{E}[Y_i(1)|X = X_i]$ denote the predicted value for individual i .

- Get $\hat{E}[Y_i(1)|X = X_i]$ for all i :

$$\hat{E}[Y_i(1)|X = X_i] = \hat{\alpha}_1 + \hat{\beta}_{11}X_{1i} + \hat{\beta}_{12}X_{2i} + \dots + \hat{\beta}_{1K}X_{Ki}$$

- Intuition: We estimate the conditional mean $E[Y(1)|X = X_i]$ for *all* individuals in our sample (controls and treated), since we want to estimate the *ATE*.

5. Using the equation $ATE = E_X[E[Y(1)|X = x] - E[Y(0)|X = x]]$ in (3), and the predicted values $\hat{E}[Y_i(0)|X = X_i]$ and $\hat{E}[Y_i(1)|X = X_i]$ we obtained in steps 2 and 4, we obtain our estimate of the ATE as:

$$\widehat{ATE} = \frac{1}{n} \sum_{i=1}^n \left[\widehat{E}[Y_i(1)|X = X_i] - \widehat{E}[Y_i(0)|X = X_i] \right] \quad (10)$$

Intuition about the estimator of the ATE in (10):

- The estimator of the ATE in (10) is just a “sample average” version of the result using population means in equation (3). We just replaced the true means in equation (3) by their estimates:
 - We use the predicted values $\widehat{E}[Y_i(0)|X = X_i]$ and $\widehat{E}[Y_i(1)|X = X_i]$ instead of the population/true expectations $E[Y_i(0)|X = X_i]$ and $E[Y_i(1)|X = X_i]$.
 - * OLS assumes that it is possible to model the conditional expectations $E[Y_i(0)|X = X_i]$ and $E[Y_i(1)|X = X_i]$ using a linear model.
 - Then, we use the sample average $\frac{1}{n} \sum_{i=1}^n [\cdot]$ in place of the population expectation $E_X[\cdot]$ to average over all the individuals in our sample.
- We can think of the term being averaged in equation (10) as an estimate of the $ATE(X_i)$, say, the average treatment effect for individuals with $X = X_i$. To see this, note that we can write:

$$\begin{aligned} \widehat{E}[Y_i(1)|X = X_i] - \widehat{E}[Y_i(0)|X = X_i] &= \widehat{E}[Y_i(1) - Y_i(0)|X = X_i] \\ &= \widehat{ATE}(X_i) \end{aligned}$$

Therefore, we can also write equation (10) as:

$$\widehat{ATE} = \frac{1}{n} \sum_{i=1}^n \widehat{ATE}(X_i)$$

Note that this is the same idea behind our result for the ATE in equation (3). Hence, we can think of our estimator of the ATE in equation (10) as first estimating the $ATE(X_i)$ at all points X_i (say, at the values of the covariates for each of the individuals in our sample), and then averaging all of them.¹

¹As before, we use $\widehat{ATE}(X_i)$ instead of $ATE(x)$ from equation (3), and $\frac{1}{n} \sum_{i=1}^n [\cdot]$ instead of $E_X[\cdot]$. Note that the sample average $\frac{1}{n} \sum_{i=1}^n [\cdot]$ is averaging over the distribution of the covariates in our entire sample to estimate the ATE . As usual, given a large enough random sample, the distribution of covariates in our sample will be similar to the distribution of covariates in the population, so the sample mean $\frac{1}{n} \sum_{i=1}^n [\cdot]$ is a good estimator of the population mean $E_X[\cdot]$.

Alternative Intuition for the Five-Step Approach to Estimate the ATE

How is the five-step approach solving the Fundamental Problem of Causal Inference (FPCI)?

Ideally, if we could observe both $Y(0)$ and $Y(1)$ for the individuals in our data, we would estimate the ATE as:

However, what we actually *observe* in our data is:

Idea: Under unconfoundedness, overlap and assuming the conditional means of $Y(0)$ and $Y(1)$ are linear, *impute the missing potential outcomes using OLS Models*.

If $T_i = 0 \implies$: use OLS model for $E[Y(1)|X] = E[Y|T = 1, X = x]$ to predict $Y_i(1)$ using i 's X 's:

$$\implies \hat{Y}_i(1) = \hat{E}[Y_i(1)|X = X_i]$$

If $T_i = 1 \implies$: use OLS model for $E[Y(0)|X] = E[Y|T = 0, X = x]$ to predict $Y_i(0)$ using i 's X 's:

$$\implies \hat{Y}_i(0) = \hat{E}[Y_i(0)|X = X_i]$$

Note:

- Note that in the five-step procedure above we used the predicted values $\hat{Y}_i(0) = \hat{E}[Y_i(0)|X = X_i]$ and $\hat{Y}_i(1) = \hat{E}[Y_i(1)|X = X_i]$ for *all* individuals in our sample, while in the example/explanation above we used $\hat{Y}_i(0)$ only for the treated individuals and $\hat{Y}_i(1)$ only for the control units (since we observe $Y_i(0)$ for controls and $Y_i(1)$ for the treated).
- Importantly, the estimated ATE in equation (10) will be exactly the same if we use $Y_i = \underbrace{Y(0)}_{\text{Observed}}$ or $\underbrace{\hat{Y}_i(0)}_{\text{Predicted}}$ for units with $T = 0$, and $Y_i = \underbrace{Y(1)}_{\text{Observed}}$ or $\underbrace{\hat{Y}_i(1)}_{\text{Predicted}}$ for units with $T = 1$.

This is because for estimation of the ATE we take the average (see equation (10)), and a property of OLS is that the average of the predicted outcomes *equals* the average of the actual outcomes, or, for an outcome Y , that:

$$\frac{1}{n} \sum_{i=1}^n Y_i = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i$$

Thus, for calculation of the estimator of the ATE in (10), it does not matter if we use the predicted or the actual outcomes, as it would be the same.

2.2.2 A Better Way to Compute the Five-Step Estimator of the ATE in (10)

A drawback of the five-step approach discussed above is that it is not straightforward to obtain the standard error of our estimator of the ATE in (10).

- Using the standard error from computing the sample average in equation (10) would not be appropriate because it ignores the fact that $\hat{Y}_i(0)$ and $\hat{Y}_i(1)$ are *predicted* (rather than the *true* values of $Y(1)$ and $Y(0)$).
- Hence, the standard errors resulting from that sample average could be severely underestimated (i.e., lower than the true standard errors).
 - Recall from your econometrics classes that the variance of the *predicted* outcomes (or *fitted* values) is less than the variance of your *actual* outcome.

A better (and simpler) way to calculate the five-step approach estimator of the ATE in (10) and, importantly, obtain its standard error, is by running the following regression:

$$Y_i = \alpha + \tau T_i + \beta_1 X_{1i} + \dots + \beta_K X_{Ki} + \gamma_1 T_i (X_{1i} - \bar{X}_1) + \dots + \gamma_K T_i (X_{Ki} - \bar{X}_K) + \varepsilon_i, \quad (11)$$

where $\bar{X}_j$ equals the sample mean of X_j , with $j = 1, 2, \dots, K$. For example, if X_1 is age, $\bar{X}_1$ would be the average age in our *entire* sample.

- In short, equation (11) is an OLS of the outcome Y_i on T_i , X_i , and $T_i \cdot (X_i - \bar{X})$.
- In this case, the estimated coefficient on T_i (i.e., τ) is *numerically equal* to our estimator of the ATE in equation (10). In other words, it can be shown that:

$$\hat{\tau} = \widehat{ATE} = \frac{1}{n} \sum_{i=1}^n \left[\hat{E}[Y_i(1)|X = X_i] - \hat{E}[Y_i(0)|X = X_i] \right]$$

- An advantage of this way to compute $\widehat{ATE}$ over the five-step approach discussed in section 2.2.1, is that from the OLS in equation (11) we directly obtain the standard error of $\hat{\tau} = \widehat{ATE}$.

3 When does an OLS of Y on T and X give us an Estimate of the ATE ?

A regression that (unfortunately) is commonly used to estimate the ATE is an OLS of the outcome Y on T and the covariates X , or:

$$Y_i = \alpha + \tau T_i + \beta_1 X_{1i} + \dots + \beta_K X_{Ki} + \varepsilon_i \quad (12)$$

It is not uncommon for people to interpret the estimated coefficient on T , τ , as an estimate of the ATE . In this section, we ask, under what assumptions can we say that τ in the OLS in (12) equals the ATE (i.e., $\tau = ATE$)?

- In other words, under what assumptions can we interpret the estimate of τ in the OLS in (12) as the causal ATE ? As you will see, the assumptions are even stronger than before.

We impose the following assumptions.

Assumption 3' (Linearity of the Conditional Mean of $Y(0)$). Assume that the conditional mean $E[Y(0)|X = x]$ is a *Linear* and *Additive* function of X . That is, assume that:

$$E[Y(0)|X = x] = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_K X_K, \text{ or}$$

$$Y(0) = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_K X_K + \varepsilon,$$

with $E[\varepsilon] = E[\varepsilon|X] = 0$.

Note that Assumption 3' is similar to Assumption 3, with the distinction that it imposes a linear model only for $E[Y(0)|X = x]$ (and not for $E[Y(1)|X = x]$). This is because of the following assumption (as explained below).

Assumption 4 (Constant Individual Treatment Effects). Assume that the individual causal treatment effect is the *same* for *all* individuals in the population:

$$Y_i(1) - Y_i(0) = \tau, \text{ for all } i.$$

Notes:

- In our leading example, Assumption 4 states that the effect of attending college on earnings is *the same for every individual* (e.g., it is the same for females, males, rich people, poor people, etc.).
- This is a *very strong* assumption! It requires all individuals to have the same effect.
- Assumption 4 implies that $ATE = ATT = ICE = \tau$ (Individual Causal Effect), since every individual has the *same* causal effect of T on Y .

Under Assumptions 1 (Unconfoundedness), 2 (Overlap), and 3' (Linearity of the Conditional Mean of $Y(0)$), and Assumption 4 (Constant Individual Treatment Effects), we have that $\tau = ATE$, where τ is the coefficient for T in the OLS in (12).

Why?

Assumption 4 states that the effect is constant across individuals, or $Y_i(1) - Y_i(0) = \tau$. Hence, we can write the potential outcome under treatment for individual i as:

$$Y_i(1) = Y_i(0) + \tau$$

- In words, the potential outcome under treatment for individual i equals her potential outcome under control plus the effect of T on Y , which is assumed to be the same for all individuals.
- Note that this is why a model for $Y(1)$ is not needed in Assumption 3' if Assumption 4 is imposed: Under Assumption 4, the model for $Y(1)$ is the same as for $Y(0)$ plus a constant τ (the effect), say, $E[Y(1)|X = x] = E[Y(0)|X = x] + \tau$
- Assumption 4 implies that the β 's in equation (4) from Assumption 3 equal the β 's in equation (5) from Assumption 3, which further imply $\alpha_1 = \alpha_0 + \tau$.

Recall that the observed outcome Y_i can be written as:

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0) = \begin{cases} \text{if } T_i = 1 \implies Y_i = Y_i(1) \\ \text{if } T_i = 0 \implies Y_i = Y_i(0) \end{cases}.$$

Then, using the fact that $Y_i(1) = Y_i(0) + \tau$, we can write the *observed* outcome as:

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$$

$$Y_i = T_i [Y_i(0) + \tau] + (1 - T_i) Y_i(0)$$

$$Y_i = T_i Y_i(0) + \tau T_i + Y_i(0) - T_i Y_i(0)$$

$$Y_i = \tau T_i + Y_i(0)$$

Then, using our model for $Y(0)$ in Assumption 3', we obtain:

$$Y_i = \tau T_i + \underbrace{\alpha + \beta_1 X_{1i} + \dots + \beta_K X_{Ki}}_{Y_i(0)} + \varepsilon_i$$

This is the same OLS as in equation (12)! Therefore, $\tau = ATE$ under Assumptions 1, 2, 3' and 4.

Notes:

- The five-step approach in section 2.2.1 imposed Assumptions 1, 2, and 3 to estimate the *ATE*, while the OLS of Y on T and X in equation (12) imposes Assumptions 1, 2, 3', and 4. In principle, given how strong the constant treatment effect assumption (Assumption 4) is, the five-step approach seems preferable. The five-step approach is *more flexible* as it allows for heterogeneous effects; that is, it allows for individuals to have different treatment effects.

- To gain some intuition on why the five-step approach allows for heterogeneous effects while the OLS in equation (12) does not, take the derivative of the OLS in equation (12) with respect to T_i :

$$Y_i = \beta_0 + \tau T_i + \beta_1 X_1 + \dots + \beta_K X_K + \varepsilon_i$$

to obtain:

$$\frac{dY_i}{dT_i} = \tau$$

Thus, the effect (τ) is constant for every individual i .

Now, if we take the same derivative of the OLS in equation (11), which is used to obtain our estimate of the *ATE* in the five-step approach and is given by:

$$Y_i = \alpha + \tau T_i + \beta_1 X_{1i} + \dots + \beta_K X_{Ki} + \gamma_1 T_i (X_{1i} - \bar{X}_1) + \dots + \gamma_K T_i (X_{Ki} - \bar{X}_K) + \varepsilon_i,$$

we obtain:

$$\frac{dY_i}{dT_i} = \tau + \gamma_1 (X_{1i} - \bar{X}_1) + \dots + \gamma_K (X_{Ki} - \bar{X}_K) \quad (13)$$

Therefore, in this case the effect varies across individuals. In particular, in this case it is a function of the covariates.

- * Note that even though the *individual* effects vary across individuals in this case, the *average* effect, or *ATE*, is still τ . The reason is that the average of the derivative in (13) equals τ , since the *average* of the rest of the terms equals zero because, for any given covariate X , we have that:

$$\frac{1}{n} \sum_{i=1}^n [X_i - \bar{X}] = 0$$

- *Final Note for General Knowledge (i.e., not for this course).* In principle, the assumption of a constant individual treatment effect (Assumption 4) used to interpret τ in the OLS in (12) as the *ATE* is *sufficient* but *not necessary*. In other words, it is possible to interpret τ in the OLS in (12) as the *ATE* under slightly weaker (but still very strong) assumptions.
 - For example, a recent paper showed that it is possible to interpret τ in the OLS in (12) as the *ATE* under Assumption 1, 2, 3, and the assumption that the average treatment effect on the treated (*ATT*) *equals* the average treatment effect on the not treated (i.e., the controls) (or *ATNT*).

Assuming that the average effect of the treated is equal to the average effect of the controls is not as strong as assuming that every individual has the same treatment effect, but it is still very strong.

In our leading example, it imposes that the average effect of attending college on earnings for those who attend college is exactly the same as the average effect for those who do not attend college (and as we discussed, those two groups are very different).

- For the purposes of this class (e.g., for the assignments), we will maintain Assumptions 3' and 4, such that *we interpret τ in the OLS in (12) as the ATE under Assumptions 1, 2, 3' and 4.* (This note was just to let you know that there are other assumptions which are weaker, but still very strong, to interpret τ in (12) as the *ATE*.)

4 Can We Blindly Trust OLS Estimates of the *ATE*?

Key paper: LaLonde (1986).

- He considered the NSW (National Supported Work Demonstration) Program Social Experiment from Problem Set 1.
- We know the estimate of the *ATE* on future earnings was \$886. This is the most credible estimate of the causal average effect of the NSW, as it came from a randomized experiment.
- LaLonde's (1986) Cool Idea: Pretend we do not have the experimental control group (i.e. the one from the experiment) and estimate the effect of the NSW program as an econometrician (in the 1980's) would do (e.g., use OLS Y on T and X , among other methods).

- What would be our estimate? Do we get an estimate close to $\widehat{ATE} = \$886$?
- LaLonde's paper:
 1. He got a sample of workers from other (1978) surveys:
 - CPS: Current Population Survey
 - PSID: Panel Study of Income Dynamics
 2. He applied many econometrics methods (from the 1980's), including OLS.
- What were his results?

See Problem Set 2 and the paper for details, but in summary:

- OLS estimates performed very poorly.
- He could not replicate experimental results.
- The results were very sensitive to the model used.
- LaLonde's paper generated a lot of interest in estimation of causal effects using both experimental and non-experimental methods.

5 When Does OLS Work Well to Adjust for X under Assumptions 1 and 2?

In what follows, we assume that Assumptions 1 (Unconfoundedness) and 2 (Overlap) hold.

Under Unconfoundedness and Overlap, when does OLS work well to adjust for covariates to estimate the ATE ?

In general, *OLS works well if*:

1. $E[Y(0)|X = x]$ and $E[Y(1)|X = x]$ are indeed *linear* functions of the covariates X ; that is, if Assumption 3 holds. (Note that, in general, this may be unlikely.)
2. Or, if $\bar{X}_1 \approx \bar{X}_0$; that is, if the average characteristics of the treatment and control groups are (relatively) close to each other. In other words, if the treatment and control groups are not very different.

Notes:

- If T is Randomly Assigned (as in an experiment) $\implies \bar{X}_1 \approx \bar{X}_0$, so OLS works well to adjust for minor imbalances in the covariates (e.g., see Problem Set 1).
- If $\bar{X}_1$ is very different from $\bar{X}_0 \implies$ OLS can do *very* poorly, *even if unconfoundedness holds!*

Intuition:

- Remember the intuition of the OLS approach:
 - To model $Y(0)$, we estimate an OLS for $E[Y(0)|X = x] = E[Y|T = 0, X = x]$ using controls, and use this model to predict $Y(0)$ for the treated.
 - To model $Y(1)$, we estimate an OLS for $E[Y(1)|X = x] = E[Y|T = 1, X = x]$ using the treated, and use this to predict $Y(1)$ for the controls.
- Even if Assumption 3 does not hold, we can think of the OLS models above as *Linear Approximations* to the true models for $E[Y|T = 0, X = x]$ around $\bar{X}_0$, and $E[Y|T = 1, X = x]$ around $\bar{X}_1$.
- **Key:** Linear approximations are good *if changes/differences are small. Otherwise, they may be off* by a lot.
 - Therefore, if the $T = 1$ and $T = 0$ groups differ a lot (i.e. if $\bar{X}_0$ is very far from $\bar{X}_1$) $\implies$ Linear approximations may be way off.
- Example: Suppose we have *only one* continuous covariate X (e.g., I.Q.).

- Analogy: Using derivatives (more formally, a line or a Taylor expansion) to approximate a function.

Example: Approximate a function $f(x)$ at x_1 , or $f(x_1)$, using a linear approximation at x_0 (say, the line tangent to $f(x_0)$).

- In Summary: If $\overline{X}_0$ is very far from $\overline{X}_1$, then OLS may do a poor job adjusting for X (unless $E[Y|T = 0, X = x]$ and $E[Y|T = 1, X = x]$ are very linear, or exactly linear so that Assumption 3 holds).
- Final Note: New approaches try to move away from using linearity assumptions.
 - These methods adjust for X in a more *flexible* way: They *only* require Assumptions 1 and 2.
 - Some examples:
 - * Nonparametric Methods. Example: Let $E[Y|T = 0, X = x] = g_0(x)$ and $E[Y|T = 1, X = x] = g_1(x)$, for any unknown functions $g_0(x)$ and $g_1(x)$.
 - * Matching Methods (later in the course)
 - * Propensity Score (p-score) Methods (later in the course)

6 Estimation of the ATT Based on OLS

The ideas used to estimate the ATT using OLS in unconfoundedness-based designs are similar to those used to estimate the ATE in section 2. The main difference is that now we focus on the average effect *only for treated individuals* (those with $T = 1$).

As discussed in the previous notes, using the Law of Iterated Expectations in the third line below, we can write the ATT as:

$$\begin{aligned} ATT &= E[Y(1) - Y(0)|T = 1] \\ ATT &= E[Y(1)|T = 1] - E[Y(0)|T = 1] \\ \implies ATT &= E[Y|T = 1] - E_{X|T=1}[E[Y(0)|T = 1, X = x]|T = 1] \end{aligned} \quad (14)$$

As we know, for the ATT we have only one missing counterfactual: $E[Y(0)|T = 1]$.

Estimation of the ATT with OLS under unconfoundedness and overlap is based on equation (14).

How do we estimate $E[Y|T = 1]$ in (14)?

As usual, we use its sample analogue: we use the sample mean instead of the population mean $E[Y|T = 1]$. Therefore, we estimate $E[Y|T = 1]$ as:

$$\hat{E}[Y|T = 1] = \frac{\sum_{i=1}^n T_i Y_i}{\sum_{i=1}^n T_i} \quad (15)$$

- In words: We estimate $E[Y|T = 1]$ using the *sample average of the outcome for those with $T_i = 1$* in our sample.

How do we estimate the missing counterfactual $E[Y(0)|T = 1]$?

As discussed before, for the ATT we need to learn about the potential value $Y(0)$ for treated individuals. Hence, we only need to model $Y(0)$ (and *not* $Y(1)$, since $Y(1)$ is observed for treated individuals).

Therefore, we impose *Assumption 3'* (see section 3): We assume that the conditional mean $E[Y(0)|X = x]$ is a linear and additive function of X .

Using our result in (14) that $E[Y(0)|T = 1] = E_{X|T=1}[E[Y(0)|T = 1, X = x]|T = 1]$, along with unconfoundedness and overlap, we can write $E[Y(0)|T = 1]$ as:

$$E[Y(0)|T = 1] = E_{X|T=1}[E[Y(0)|T = 1, X = x]|T = 1] \quad (16)$$

$$E[Y(0)|T = 1] = E_{X|T=1}[E[Y(0)|T = 0, X = x]|T = 1]$$

$$E[Y(0)|T = 1] = E_{X|T=1}[E[Y|T = 0, X = x]|T = 1] \quad (17)$$

Under Assumption 3' (Linearity of the Conditional expectation of $Y(0)$) and using the fact that $Y(0)$ equals Y for units with $T = 0$, we can estimate the inner expectation $E[Y|T = 0, X = x]$ by running an OLS of Y on X for those with $T = 0$.

After we estimate a model for $Y(0)$ (specifically, for $E[Y|T = 0, X = x]$ in equation (17)), we can use it to predict $Y(0)$ *for the treated* (specifically, $E[Y(0)|T = 1, X = x]$ in equation (16)). Then, we can average it over the distribution of the covariates *in the treated group* to obtain the *ATT*. This last step corresponds to the outer conditional mean $E_{X|T=1}[\cdot|T = 1]$ in equation (16).

The equations above are the foundation for the following approach to estimate the *ATT*.

6.1 Three-Step Approach to Estimate the *ATT* Using OLS

As before, we assume that we have a random sample size n from our population of interest. Based on the discussion above, *under Assumptions 1 (Unconfoundedness), 2 (Overlap), and 3' (Linearity of the Conditional Mean of $Y(0)$)*, we can estimate the *ATT* using the following three-step procedure:

1. Run OLS of Y on X using units in the control group ($T = 0$).
 - Here, we get estimates of $\alpha, \beta_1, \dots, \beta_K$ in Assumption 3'.
 - Intuition: We are estimating the conditional mean of the potential outcome $Y(0)$, $E[Y(0)|X = x]$, in Assumption 3' using OLS:

$$Y(0) = \alpha + \beta_1 X_1 + \dots + \beta_K X_K + \varepsilon$$

2. Get predicted values *for the treated units* in our sample. Let the predicted value for *treated* individual i be denoted by $\hat{Y}_i(0) = \hat{E}[Y_i(0)|T_i = 1, X = X_i]$.

- Get $\hat{Y}_i(0) = \hat{E}[Y_i(0)|T_i = 1, X = X_i]$ for all *treated* individuals in our sample as:

$$\hat{Y}_i(0) = \hat{\alpha} + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_K X_K$$

3. Using the equation $E[Y(0)|T = 1] = E_{X|T=1}[E[Y(0)|T = 1, X = x]|T = 1]$ in (16), the predicted values $\hat{Y}_i(0) = \hat{E}[Y_i(0)|T_i = 1, X = X_i]$ we obtained in step 2, and our estimator of $E[Y|T = 1]$ in equation (15), we obtain our estimate of the *ATT* as:

$$\widehat{ATT} = \frac{\sum_{i=1}^n T_i Y_i}{\sum_{i=1}^n T_i} - \frac{\sum_{i=1}^n T_i \cdot \hat{E}[Y_i(0)|T_i = 1, X = X_i]}{\sum_{i=1}^n T_i} \quad (18)$$

Or, using the simpler notation that $\hat{Y}_i(0) = \hat{E}[Y_i(0)|T_i = 1, X = X_i]$:

$$\widehat{ATT} = \frac{\sum_{i=1}^n T_i Y_i}{\sum_{i=1}^n T_i} - \frac{\sum_{i=1}^n T_i \hat{Y}_i(0)}{\sum_{i=1}^n T_i} \quad (19)$$

Intuition about the estimator of the *ATT* in (18) and (19):

- In words: Our estimator of the *ATT* in (18) and (19) equals the sample average of our outcome Y_i *for those with* $T_i = 1$ (the first term) *minus* the sample average of our *predicted potential outcome* $\hat{Y}_i(0)$ *for those with* $T_i = 1$ (the second term).
- As for the *ATE*, our estimator of the *ATT* in (18) and (19) is simply a “sample average” version of the result using population means in equation (14).
 - **Importantly**, note that while our estimator of the *ATE* used sample averages over the *entire* sample (since it estimates the average effect for the *population*), our estimator of the *ATT* uses sample averages over *the treated individuals in our sample* (since it estimates the effect *for the treated*).

- Unlike the five-step approach to estimate the ATE in section 2, we cannot calculate our estimator $\widehat{ATT}$ in (18) and (19) using only one Regression. We actually need to perform the three-step approach described above to get $\widehat{ATT}$.
 - Therefore it is not straightforward to get the standard error of $\widehat{ATT}$.
 - That can be done using a procedure called “*Bootstrap*” (which is not covered in this class). Intuitively, it is calculated by using simulation methods.
- Intuition for the three-step approach to estimate the ATT :
 - As for the ATE , the key insight is that under unconfoundedness, overlap, and assuming the conditional mean of $Y(0)$ is linear, we *impute the missing potential outcomes $Y(0)$ for the treated individuals using an OLS model*.